A determinant-one counterexample to the proposed strong-(B) equality classification

Automated Math Discovery Run `fa_banach_001`, lane 8

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Result. The equality characterization in Conjecture 4.1 of Artstein-Avidan–Katzin is false. There is a centrally symmetric unconditional convex body $K \subset \mathbb{R}^3$ and a nonzero, trace-zero, nonscalar diagonal matrix Λ for which $\mathrm{vol}(e^{s\Lambda}K \cap B_2^3)$ is constant on a nontrivial interval, although neither $K \subset B_2^3$ nor $B_2^3 \subset K$.

1 Source conjecture

Conjecture 4.1 of [1] asserts that, for a convex body $K \subset \mathbb{R}^n$ and diagonal Λ,

$$\phi(t) = \mathrm{vol}_n(e^{t\Lambda}K \cap B_2^n)$$

is log-concave. In particular it proposes

$$\mathrm{vol}_n(e^{t\Lambda/2}K \cap B_2^n)^2 \geq \mathrm{vol}_n(e^{t\Lambda}K \cap B_2^n)\ \mathrm{vol}_n(K \cap B_2^n). \tag{1}$$

It further states that equality occurs if and only if $K \subset B_2^n$, or $B_2^n \subset K$, or $\Lambda = \lambda I_n$. The paper uses this rigidity clause, with $\mathrm{tr}\,\Lambda = 0$, to deduce uniqueness of a maximum intersection ellipsoid in the intermediate-radius regime.

2 Idea of the counterexample

Let one coordinate of a box cut the unit ball and make the other coordinate widths larger than the ball. A diagonal deformation supported only on the oversized coordinates is invisible to intersection with the ball for as long as those widths remain at least one. Two such redundant coordinates can expand and contract reciprocally, making the deformation nontrivial and trace zero while leaving the intersection exactly fixed.

3 The counterexample

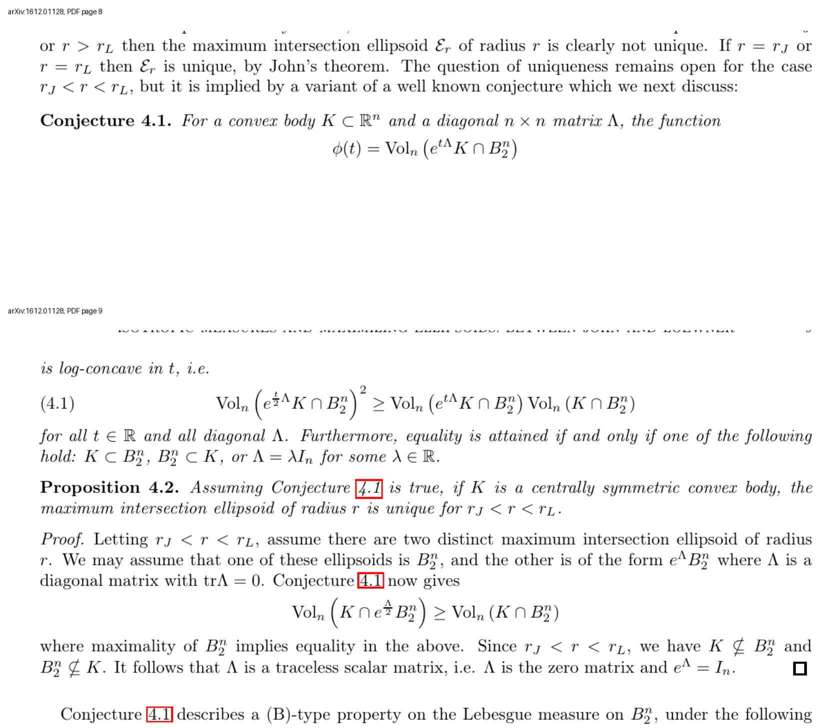
Theorem 1. *Set*

$$K = [-\tfrac{1}{2}, \tfrac{1}{2}] \times [-2, 2] \times [-3, 3] \quad \textit{and} \quad \Lambda = \mathrm{diag}(0, 1, -1).$$

Then, for every $s \in [0, 1]$,

$$e^{s\Lambda}K \cap B_2^3 = \{x \in B_2^3 : |x_1| \leq \tfrac{1}{2}\}. \tag{2}$$

Consequently (1) *is an equality at $t = 1$, although none of its three proposed equality conditions holds. Moreover* $\mathrm{tr}\,\Lambda = 0$.



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or $r > r_L$ then the maximum intersection ellipsoid $\mathcal{E}_r$ of radius r is clearly not unique. If $r = r_J$ or $r = r_L$ then $\mathcal{E}_r$ is unique, by John's theorem. The question of uniqueness remains open for the case $r_J < r < r_L$, but it is implied by a variant of a well known conjecture which we next discuss:

Conjecture 4.1. *For a convex body $K \subset \mathbb{R}^n$ and a diagonal $n \times n$ matrix Λ, the function*

$$\phi(t) = \mathrm{Vol}_n\left(e^{t\Lambda}K \cap B_2^n\right)$$

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is log-concave in t, i.e.

$$\mathrm{Vol}_n\left(e^{\frac{t}{2}\Lambda}K \cap B_2^n\right)^2 \geq \mathrm{Vol}_n\left(e^{t\Lambda}K \cap B_2^n\right)\mathrm{Vol}_n\left(K \cap B_2^n\right) \tag{4.1}$$

for all $t \in \mathbb{R}$ and all diagonal Λ. Furthermore, equality is attained if and only if one of the following hold: $K \subset B_2^n$, $B_2^n \subset K$, or $\Lambda = \lambda I_n$ for some $\lambda \in \mathbb{R}$.

Proposition 4.2. *Assuming Conjecture 4.1 is true, if K is a centrally symmetric convex body, the maximum intersection ellipsoid of radius r is unique for $r_J < r < r_L$.*

Proof. Letting $r_J < r < r_L$, assume there are two distinct maximum intersection ellipsoid of radius r. We may assume that one of these ellipsoids is B_2^n, and the other is of the form $e^{\Lambda}B_2^n$ where Λ is a diagonal matrix with $\mathrm{tr}\Lambda = 0$. Conjecture 4.1 now gives

$$\mathrm{Vol}_n\left(K \cap e^{\frac{\Lambda}{2}}B_2^n\right) \geq \mathrm{Vol}_n\left(K \cap B_2^n\right)$$

where maximality of B_2^n implies equality in the above. Since $r_J < r < r_L$, we have $K \not\subseteq B_2^n$ and $B_2^n \not\subseteq K$. It follows that Λ is a traceless scalar matrix, i.e. Λ is the zero matrix and $e^{\Lambda} = I_n$. □

Conjecture 4.1 describes a (B)-type property on the Lebesgue measure on B_2^n, under the following

Figure 1: Conjecture 4.1 and Proposition 4.2 in arXiv:1612.01128, PDF pages 8–9.

Proof. For $s \in [0, 1]$,

$$e^{s\Lambda}K = [-\tfrac{1}{2}, \tfrac{1}{2}] \times [-2e^s, 2e^s] \times [-3e^{-s}, 3e^{-s}].$$

The second half-width satisfies $2e^s \geq 2 > 1$, and the third satisfies $3e^{-s} \geq 3/e > 1$. Since every $x \in B_2^3$ has $|x_2|, |x_3| \leq 1$, those two box constraints are redundant inside the ball. This proves (2).

Writing $a = 1/2$ and integrating the disk cross-sections gives the common volume

$$\phi(s) = \int_{-a}^{a} \pi(1 - u^2)\,du = 2\pi\left(a - \frac{a^3}{3}\right) = \frac{11\pi}{12}.$$

Thus $\phi(1/2)^2 = \phi(1)\phi(0)$.

The point $(0, 2, 0)$ belongs to $K \setminus B_2^3$, while $(1, 0, 0)$ belongs to $B_2^3 \setminus K$. Finally, Λ has three unequal diagonal entries, so it is not scalar, while $\mathrm{tr}\,\Lambda = 0 + 1 - 1 = 0$. □

4 A robust family of equality cases

Proposition 2. *Let $0 < a < 1$, let $I \subset \mathbb{R}$ be an interval, and put*

$$K = [-a, a] \times \prod_{j=2}^{n} [-b_j, b_j], \qquad \Lambda = \mathrm{diag}(0, \lambda_2, \ldots, \lambda_n).$$

If $b_j e^{s\lambda_j} \geq 1$ for every $s \in I$ and every $j \geq 2$, then

$$e^{s\Lambda}K \cap B_2^n = \{x \in B_2^n : |x_1| \leq a\} \qquad (s \in I).$$

In particular, in every dimension $n \geq 3$ one may choose a nonzero trace-zero nonscalar Λ and obtain equality on any prescribed compact interval after taking the b_j's sufficiently large.

Proof. Inside B_2^n, every coordinate has absolute value at most one. The hypothesis therefore makes all constraints except $|x_1| \le a$ redundant, proving the set identity. For the final assertion, take two indices $j, k \ge 2$, set $\lambda_j = c$, $\lambda_k = -c$ with $c \ne 0$, and set the remaining eigenvalues to zero. On a compact interval, finite b_j, b_k can dominate the largest reciprocal scaling. $\square$

This family shows that the failure is structural. It is not caused by choosing $t = 0$, by a scalar dilation, by non-central symmetry, or by a volume-changing deformation. The intersection set itself, not merely its volume, remains fixed on a whole interval.

5 Why this does not yet settle ellipsoid uniqueness

Because $\operatorname{tr} \Lambda = 0$, change of variables gives

$$\operatorname{vol}(e^{s\Lambda} K \cap B_2^3) = \operatorname{vol}(K \cap e^{-s\Lambda} B_2^3).$$

Thus a continuum of distinct equal-volume ellipsoids has the same overlap with K. To refute uniqueness of the *maximizing* ellipsoid, one would still have to prove that this common value is the global maximum. It is not so for the displayed example.

Indeed, use the source's first-variation formula at the unit ball with $A = \operatorname{diag}(-2, 1, 1)$. On the sphere, $S^2 \cap K = \{x \in S^2 : |x_1| \le a\}$, $a = 1/2$, because the other box constraints are redundant. Since spherical area disintegrates as $dS = 2\pi\, du$ with $u = x_1$,

$$\begin{aligned} \frac{d}{dt}\bigg|_{t=0} \operatorname{vol}(K \cap e^{tA} B_2^3) &= \int_{S^2 \cap K} \langle x, Ax \rangle \, dS \\ &= 4\pi a - 3\left(\frac{4\pi a^3}{3}\right) = 4\pi a(1 - a^2) > 0. \end{aligned}$$

So the unit ball is not even a local maximizer. This closes the most direct upgrade route without overstating the result: the equality rigidity in Conjecture 4.1 is false, while the log-concavity inequality and the paper's intermediate-radius uniqueness question remain open here.

6 Verification and novelty notes

The proof is an exact coordinate calculation. The accompanying script checks all endpoint widths on a 1001-point grid, the exact cross-sectional volume, and the positive first variation; the grid is only a transcription check, since monotonicity proves the bounds analytically.

A bounded official-arXiv search through 11 August 2026 used arXiv id 1612.01128 and combinations of "strong B equality", "uniform measure ball equality", and "maximal intersection ellipsoid uniqueness". It found the source and related strong-B literature but no later answer to this exact equality claim. This is not an exhaustive literature review.

References

[1] S. Artstein-Avidan and D. Katzin, *Isotropic Measures and Maximizing Ellipsoids: Between John and Loewner*, arXiv:1612.01128.